

Hardware Requirements Specification

Sentinel Version 1.0

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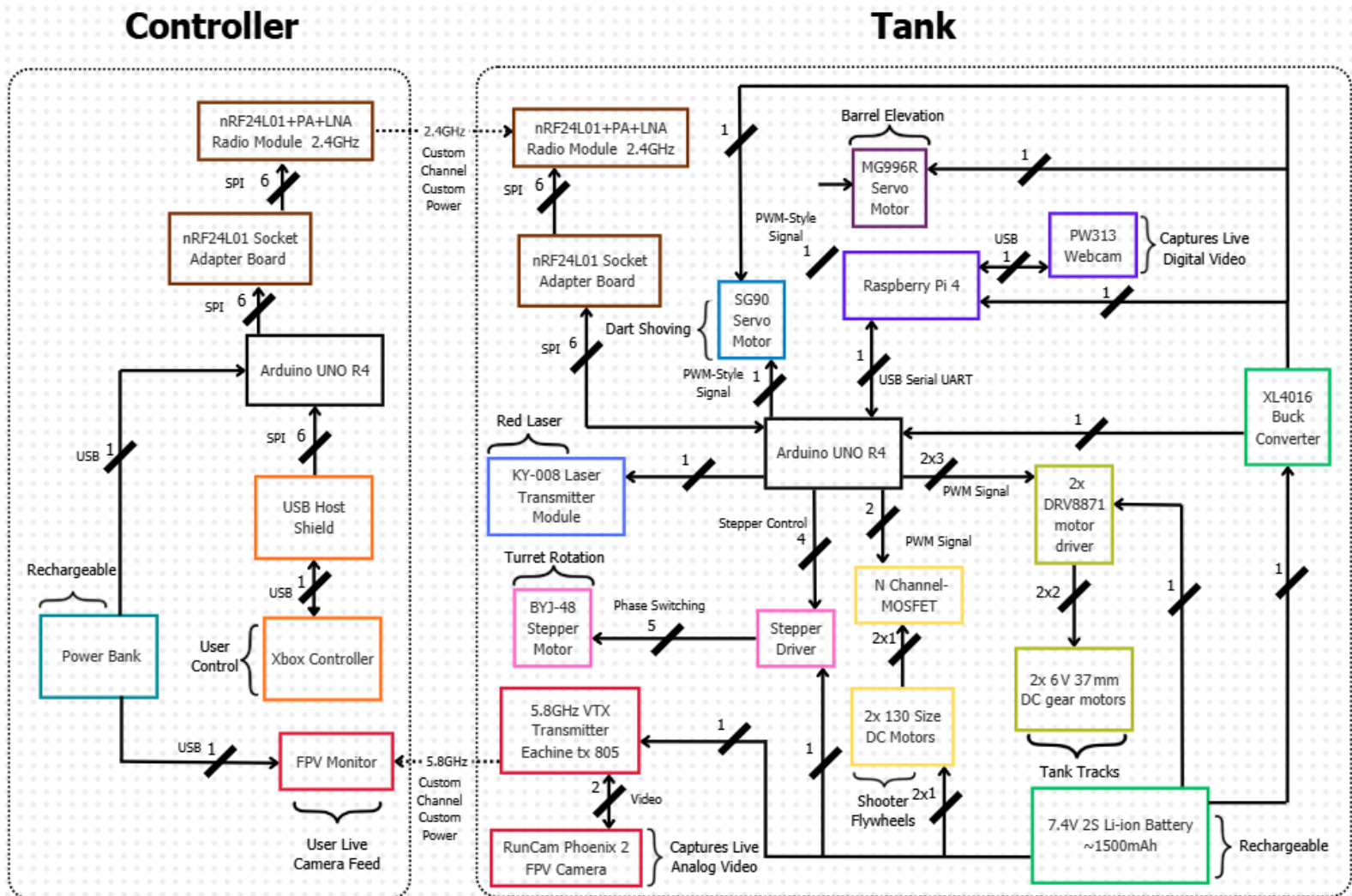
TEJ4M, Mr. Roller

1. Executive Summary

1.1 Project Overview

The Sentinel is a robotic toy tank that combines fun, engineering, and usefulness in one project. By letting kids or any bored adults get ahold of this engineering masterpiece, they can have fun or malice as required. The tank, which can shoot nerf darts at people, has many features like any real tank, including a rotational turret and tank tracks, and can be remotely controlled by a user who can watch a video stream connected to the tank and control it via a gaming controller.

High-Level Block Diagram of the Sentinel



1.2 Purpose and Scope of this Specification

The purpose behind this project is to create a fun, meaningful toy that can bring joy to any customer interested in the capabilities of a hobby robot tank. As the designer of the Sentinel, we seek to build and assemble a functional robot that our clients and our manager will be satisfied with. Clients from the toy hobby RC and the toy manufacturing industry require a cheap toy that can shoot toy darts at human targets, with various other functional requirements, such as being able to stream a live video feed to a controller far away. Clients also require that this project meet most of the curriculum expectations of the Ontario TEJ4M course, covering many strands like circuitry and networking. Additionally, the manager requires an efficient assembly with cheap costs, meeting deadlines as outlined in the PDD. Specifically, this project shall be completed within a bit more than two months, with planning and delivery taking an extra two months in addition to this assembly period.

2. Product Description

The Sentinel is a hobby remote-controlled robotic tank system designed to emulate a real tank and its features while being a fun toy to play with. The Sentinel shall resemble a moving turret, made of a plastic frame with movable parts. The tank shall be controlled by a remote gaming controller, moving its tank tracks around, rotating a turret, and shooting Nerf darts. Moreover, the tank will stream a live video feed from the tank to the user through an FPV drone monitor, so the user can effectively control it from far away, out of sight. The Sentinel shall have many features. It can reload using magazines; it can use vision processing to autonomously identify human targets, aiming towards them; and it can even shoot Nerf darts rapidly in succession—automatically. Likewise, the tank will also contain many electronics and interfaces—both the controller and tank shall be largely Arduino-based but will also contain a Raspberry Pi 4, rechargeable batteries, radio modules, a drone camera and monitor, powerful motors, and more.

2.1 Product Context

As a partially autonomous, controllable tank, the Sentinel does not regularly interact or interface with any external third-party software or hardware: it is a self-contained, independent system that can be controlled by users. However, the fabrication of the Sentinel uses many outsourced parts and software to run the tank. For example, the tank shoots Nerf darts as its projectile, and the tank uses Arduino and Raspberry Pi microcontrollers to run software and control input and output devices. Additionally, the tank requires many consumer products in assembly, such as an Xbox controller, a Runcam Phoenix 2 drone camera, and a SoloGood drone monitor to control the tank.

2.2 User Characteristics

Ideally, the general customer would be any teenager or interested RC hobbyist who would be interested in playing with and using a toy remote-controlled tank. These hobbyists would be adult engineers, teenagers, or anyone trying to get in the field and learning about robotics or remote control. However, because the project's user interface does not require any technical expertise, kids and almost any client simply looking for a fun time can use this product, even if they cannot understand or learn about the electronics and theory behind the Sentinel.

2.3 Assumptions/Risks

The following assumptions describe conditions expected to be true for the system during development and operation. These assumptions help define the environment and limitations under which the tank is designed to function

Assumption 1: It is assumed that the tank will operate in an indoor or well-controlled outdoor environment with minimal obstacles and hazards.

Assumption 2: It is assumed that the communication signal between the controller and the tank will remain within the intended operating range.

Assumption 3: It is assumed that the tank will not be exposed to water, extreme temperatures, or harsh environmental conditions.

Assumption 4: It is assumed that the operator has basic knowledge of how to control the tank.

Assumption 5: It is assumed that the components and wireless communication systems will perform according to their documented specifications. For example, the nRF24L01 + PA + LNA 2.4 GHz modules are expected to achieve a significant portion of their advertised maximum line-of-sight communication range under low-interference conditions.

The following risks describe potential issues that could affect the development or performance of the tank. These risks may impact progress, system reliability, or overall functionality if they occur.

Risk 1: Some electronic components that require purchase (such as the high-torque DC motors or USB host shield) may be unavailable or delayed in shipping.

Risk 2: The motors may not provide sufficient torque to move the tank if the final system weight is greater than expected.

Risk 3: Wireless communication may experience interference from other devices operating on similar frequencies.

Risk 4: Mechanical components such as motors, gears, or tracks may wear out or fail during extended use.

Risk 5: The 3D printed structural parts may not be strong enough and could crack or deform under stress during operation.

Risk 6: The system may experience overheating if motors, regulators, or other electronic components draw high current for extended periods.

2.4 Constraints

The following constraints define the limitations imposed on the design, components, and operation of the tank. These constraints must be considered during development to ensure the system functions as intended.

Constraint 1: The weight of the components on the tank must remain low enough so that the drive motors can move the tank efficiently, limiting the total number of electronic components and batteries

that can be used.

Constraint 2: The physical size of the chassis limits the space available for electronic components, requiring careful layout and compact component selection.

Constraint 3: The tank must operate using a battery power source, which restricts the total available power to electronic components and motors.

Constraint 4: The system uses an Arduino UNO R4 WiFi, which limits available memory, processing speed, and the number of I/O pins for controlling components.

Constraint 5: The system must interface with external hardware components (motor drivers, sensors, etc.), which restricts the types of communication interfaces and protocols that can be used.

2.5 Dependencies

The following dependencies identify components or modules that must function correctly or be completed before other parts of the system can operate or be tested.

Dependency 1: The tank's onboard electronics depend on the battery to supply a stable voltage.

Dependency 2: The firing mechanism depends on the flywheel motor control circuitry to operate correctly.

Dependency 3: The video transmission system depends on the RunCam Phoenix 2 FPV camera and the 5.8 GHz video transmitter functioning properly.

Dependency 4: The power distribution system must be implemented before the electronics and motors can be safely tested.

Dependency 5: The motor control system must be completed before the drive system can be tested.

Dependency 6: The camera and transmitter module must be installed before the video transmission system can be tested.

3. Requirements

3.1 Functional Requirements

Describes the functions the product shall be able to perform

Req#	Requirement	Comments	Date Rvwd	Customer Reviewed / Approved
HRS-3.1.1	The controller shall permit the user to remotely control the tank's movements and features within an indoor range of at least 50 m	Control should be reliable: low latency, correct, long range, and without frequent disconnection/distortions		Mr. Roller
HRS-3.1.2	The controller shall provide a live video feed from the tank to a user within an indoor range of at least 50 m	Video should be reliable: low latency, correct, long range, and without frequent disconnection/distortions		Mr. Roller
HRS-3.1.3	The tank shall be able to move at different speeds and directions: it shall be able to go forward and backward, along with turning left and right by pivoting with tank drive mechanics.	Tank drive mechanics: turning and driving by controlling either left or right sides of the tank treads as separate units. E.g., turn left on the spot by moving left tracks backwards and right tracks forwards at full speeds		Mr. Roller
HRS-3.1.4	The tank shall be able to fire a Nerf foam dart at a distance of at least 10 m away from the tank.			Mr. Roller
HRS-3.1.5	The tank shall have a capacity of at least 10 Nerf darts.			Mr. Roller
HRS-3.1.6	The tank shall be able to fire at a maximum fire rate of at least 1 dart per second			Mr. Roller
HRS-3.1.7	The tank's turret shall be able to rotate continuously, past 360° of rotation	Continuous, infinite rotation		Mr. Roller

HRS-3.1.8	The tank's turret shall be able to elevate at least 30° upwards from the base position	Base elevation could be below the horizontal (E.g. range of motion from -15° to 15°)		Mr. Roller
HRS-3.1.9	The tank shall include a toggleable red laser beam that indicates where the turret is pointing			Mr. Roller
HRS-3.1.10	The tank shall be able to independently identify human targets and enter a semi-autonomous mode that tracks targets with its turret	Barrel can move and rotate to aim at identified target independent of user control		Mr. Roller

3.2 Hardware Requirements

Describes the hardware characteristics of the product, including electronics and structural parts

Req#	Requirement	Comments	Date Rvwd	Customer Reviewed / Approved
HRS-3.2.1	The tank's outer body shall be made with 3D-printed parts with PLA	This includes the chassis, barrel, and rotating turret.		Mr. Roller
HRS-3.2.2	The tank's internal mechanisms, including but not limited to flywheels, custom gears, and shafts, shall be made with 3D-printed parts with PETG	Parts that are moving and need to withstand pressure and heat need to be printed with strong, PETG material		Mr. Roller
HRS-3.2.3	The tank shall be driven on two rubber tank treads, attached to the body, that are tensioned by free-spinning wheels and powered by gears	Wheels will form the shape and guide the treads, while the gear will drive and physically shift each tread.		Mr. Roller
HRS-3.2.4	The tank shall have a chassis base that supports the turret and treads	Houses electronic components as well		Mr. Roller
HRS-3.2.5	The tank shall have a turret on top of the chassis for firing	Includes barrel, housing for electronic components		Mr. Roller

HRS-3.2.6	The tank shall be able to accommodate a Nerf magazine, which will be inserted on the side of the tank and can be easily removed or inserted.	3D-printed custom magazine holder Magazine is spring loaded, containing Nerf darts		Mr. Roller
HRS-3.2.7	The tank shall be powered by a 7.4 V 2S Li-ion Battery			Mr. Roller
HRS-3.2.8	The tank shall use and contain an Arduino UNO R4 microcontroller. The Arduino will control and power certain input/output devices on the tank.			Mr. Roller
HRS-3.2.9	The tank shall use and contain a Raspberry Pi 4 for vision processing			Mr. Roller
HRS-3.2.10	The tank's Raspberry Pi 4 and Arduino Uno R4 microcontrollers shall be connected to a 5 V Buck Converter, which will be connected to the 7.4 V 2S Li-ion Battery.			Mr. Roller
HRS-3.2.11	The tank shall contain a 28BYJ-48 stepper motor and a ULN2003 driver board for rotating the turret			Mr. Roller
HRS-3.2.12	The tank shall contain an MG996R servo motor for changing the turret elevation			Mr. Roller
HRS-3.2.13	The tank shall contain a SG90 servo motor for pushing darts to be shot out			Mr. Roller
HRS-3.2.14	The tank shall use two DRV8871DDA DC motor drivers to drive the two 6V-rated JGB37-550 37mm motors for moving the tank tracks	Motors have a torque of at least 0.35 Nm and an RPM of around 100-300.		Mr. Roller
HRS-3.2.15	The tank shall use one MOSFET to run the flywheel 130-size DC motors	Motors shall be able to reach speeds of at least 25000 rpm within 2-3 seconds, operating continuously for ≥ 60 seconds without overheating		Mr. Roller

HRS-3.2.16	The tank shall have an nRF24L01+ PA/LNA Radio Module 2.4 GHz for receiving radio signals			Mr. Roller
HRS-3.2.17	The tank shall have a RunCam Phoenix 2 FPV Camera for analog video to the user			Mr. Roller
HRS-3.2.18	The tank shall have an Eachine TX805 5.8 GHz VTX to transmit analog video to the monitor			Mr. Roller
HRS-3.2.19	The tank shall have a mounted digital USB AverMedia PW313 Webcam to the turret that is connected to the Raspberry Pi 4 for vision processing			Mr. Roller
HRS-3.2.20	The controller shall have an Arduino Uno R4 microcontroller			Mr. Roller
HRS-3.2.21	The controller shall contain a USB Host Shield connecting the microcontroller to the Xbox controller	USB Host Shield powers and transfers USB data between Xbox controller and microcontroller		Mr. Roller
HRS-3.2.22	The controller's Arduino shall be powered by a rechargeable power bank via USB			Mr. Roller
HRS-3.2.23	The controller shall have a wireless Xbox controller for user control			Mr. Roller
HRS-3.2.24	The controller shall have an nRF24L01+ PA/LNA Radio Module 2.4 GHz for transmitting radio signals			Mr. Roller
HRS-3.2.25	The controller shall contain a drone SoloGood 4.3" FPV monitor for receiving and streaming live video			Mr. Roller
HRS-3.2.26	The controller shall contain a 3D-printed PLA casing to hold the controller's Arduino and power bank, and shall attach the monitor from HRS-3.2.25 to the controller			Mr. Roller

3.3 User Interface Requirements

Describes the characteristics of each interface between the product and its users

Req#	Requirement	Comments	Date Rvwd	Customer Reviewed / Approved
HRS-3.3.1	The tank shall be controlled by a user-operated remote Xbox controller that sends 2.4 GHz radio signals to control the tank.	Using nRF24L01+ PA/LNA radio modules, with Arduino UNO R4s , USB Host Shield, and standard Xbox wireless controllers The user can control the tank remotely		Mr. Roller
HRS-3.3.2	The tank shall stream live, analog video through wireless 5.8 GHz radio signals to an FPV monitor	Monitor is attached to the controller The user can watch the live video stream of the tank while they are controlling it		Mr. Roller
HRS-3.3.3	The tank shall be able to move its treads via joystick, radio frequency remote control	Tank drive, using an arcade drive control system where one joystick controls both tank treads Left joystick on Xbox controller		Mr. Roller
HRS-3.3.4	The tank shall be able to spin the flywheels when the user holds a trigger down on a radio frequency remote controller.	Flywheels will spin at different speeds based on how much the trigger is held down Left trigger on Xbox controller		Mr. Roller
HRS-3.3.5	The tank shall be able to shoot a dart when the user presses a button on a radio frequency remote controller. If the user holds the button, the tank shall be able to continuously fire darts. This feature will only work if flywheels are activated as outlined in HRS-3.3.4	Right trigger on Xbox controller		Mr. Roller

HRS-3.3.6	The turret on the tank shall be able to rotate when the user moves the joystick of a radio frequency remote control left and right. The barrel on the tank shall be able to move up and down when the user moves the same joystick up and down.	Left/right yaw movement and up/down pitch movement on same joystick Right joystick on Xbox controller		Mr. Roller
HRS-3.3.7	The tank shall be able to toggle a laser on or off with a radio frequency remote controller. The laser should indicate where the dart will hit.	Face button X on the Xbox controller		Mr. Roller
HRS-3.3.8	The tank shall be able to enter and exit an autonomous mode once a user toggles a button on a radio frequency remote controller. The autonomous mode allows the tank to move and aim at targets independent of user control, but it will not be able to fire.	Face button A on the Xbox controller		Mr. Roller
HRS-3.3.9	The tank shall automatically power off and cease movement if it loses communication with the remote controller	If signals become distorted or the tank stops receiving signals, such as if something breaks or the tank goes out of range, the tank stops all movements		Mr. Roller
HRS-3.3.10	The tank shall have a physical on/off switch that allows users to power/start the tank and power off the tank.	Switch to connect or disconnect battery		Mr. Roller
HRS-3.3.11	The tank shall be able to fire Nerf darts at users and non-user targets and hit them	Physical user-system interaction		Mr. Roller

3.4 Hardware Interfaces

Describes the internal and external communication within the product and external systems

Req#	Requirement	Comments	Date Rvwd	Customer Reviewed / Approved
HRS-3.4.1	The tank's Arduino microcontroller shall drive both DRV8871DDA DC motor drivers with a PWM signal, which in turn will drive 2 6V-rated JGB37-550 37mm motors connected to gears, which will drive the tank treads			Mr. Roller
HRS-3.4.2	The tank's Arduino microcontroller shall drive the MOSFET with a PWM signal, which will drive two 130-Size High-RPM DC Motors, which will be connected to the flywheels.			Mr. Roller
HRS-3.4.3	The tank's Arduino microcontroller shall drive the laser module with a digital output signal			Mr. Roller
HRS-3.4.4	The Arduino microcontroller shall control a stepper driver that will drive a stepper motor that will be used to rotate the turret	Logic inputs from the microcontroller into the driver determining speed, direction, enabling, step resolution Driver controls motor by phase power switching that connects 5V to ground on specified phases, individual ends of the motor coils.		Mr. Roller
HRS-3.4.5	The tank's Arduino microcontroller shall drive a servo motor with a PWM signal that will change the elevation of the barrel.			Mr. Roller
HRS-3.4.6	The tank's Arduino microcontroller shall drive a servo motor with a PWM signal that will push the dart into the flywheels			Mr. Roller
HRS-3.4.5	The tank's Arduino UNO R4 communicates with the Raspberry Pi 4 by Serial UART over USB			Mr. Roller

HRS-3.4.7	The tank shall have a digital webcam connected to a Raspberry Pi 4 via USB for vision processing.			Mr. Roller
HRS-3.4.8	The tank's Arduino microcontroller shall receive and interpret radio signals with an nRF24L01+PA+LNA radio module by SPI			Mr. Roller
HRS-3.4.9	The controller's Arduino microcontroller shall transmit and package radio signals with an nRF24L01+PA/LNA radio module by SPI			Mr. Roller
HRS-3.4.10	The tank's radio module shall receive transmitted radio signals from the controller's radio module through wireless 2.4 GHz radio signals	Radio waves will be modulated and customized data packets Antenna-to-antenna communication		Mr. Roller
HRS-3.4.11	The controller's Arduino microcontroller shall interface with a USB Host Shield by SPI	Note: both the radio module and host shield will need to share some SPI pins on the controller's microcontroller, but differ in address pins		Mr. Roller
HRS-3.4.12	The controller's USB Host Shield will connect the Xbox Controller to the microcontroller by USB			Mr. Roller
HRS3.4.13	The tank shall have an FPV camera connected to a 5.8 GHz VTX transmitter	Camera sends analog composite video to the VTX		Mr. Roller
HRS3.4.14	The tank's video transmitter will send 5.8 GHz video signal to an FPV monitor on the controller	Analog video transmitted as radio waves Monitor and VTX need to be set to be on the same channel (frequency) VTX can switch power of signal (25 mw, 200 mW, 600 mW, 800 mW), changing range, power usage, and signal integrity		Mr. Roller

4. Requirements Confirmation/Stakeholder sign-off

Include documentation of the approval or confirmation of the requirements here. For example:

Meeting Date	Attendees (name and role)	Comments
02/26/26	Designer Bryan, Designer Bruce, Manager Mr.Roller, Customer	Confirmed project proposition

APPENDIX

Appendix A. Definitions, Acronyms, and Abbreviations

Term/Abbreviation	Definition/Meaning
MOSFET	Metal-Oxide-Semiconductor Field-Effect Transistor: an electronic switch used to control high-current devices like motors.
PWM	Pulse Width Modulation: a technique to control motor speed by varying the duty cycle of a digital signal.
FPV	First-Person View: a system where a camera on the tank transmits live video to goggles or a monitor.
DC	Direct Current: a type of electrical current that flows in one direction, commonly used to power electronics
RF	Radio Frequency: electromagnetic waves used for wireless communication
GND	Ground: the reference point in an electrical circuit from which voltages are measured and a common return path for current.
VCC	Positive Voltage Supply: the positive voltage supply in a circuit that powers components like microcontrollers and sensors.
RPM	Rotations per minute: a measure of how fast a motor or wheel spins, indicating speed or rotational velocity.
PETG	Polyethylene Terephthalate Glycol: a strong thermoplastic filament used in 3D printing
PLA	Polylactic Acid: a simple thermoplastic filament used in 3D printing
VTX	Video Transmitter: a device that sends live video
SPI	Serial Peripheral Interface: a fast, full-duplex, synchronous 4-wire serial communication protocol

USB	Universal Serial Bus: the standard technology for connecting, communicating, and powering devices like computers, phones, and peripherals
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Appendix B. References

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